

Jialu Zhang

University of Waterloo
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Research Interests

My academic work is in software engineering and programming education. It focuses on methods and tools that help students and developers understand, test, and improve programs.

Work Experience

- 03/24 - Now **Tenure-track Assistant Professor**, *Electrical and Computer Engineering*, University of Waterloo, Waterloo, ON, Canada
- Duties include teaching undergraduate and graduate courses, supervising students, course preparation, and university service.
- 09/23-02/24 **Postdoc**, *Computer Science*, Yale University, New Haven, CT, USA

Education

- 08/17-05/23 **Ph.D.**, *Computer Science*, Yale University, New Haven, CT, USA
- 09/13-07/17 **B.S.**, *Computer Engineering*, Shanghai Jiao Tong University, Shanghai, Shanghai, China
- 09/10-06/13 **High School**, *Academic*, High School Affiliated to Shanghai Jiao Tong University, Shanghai, Shanghai, China
- 09/06-06/10 **Junior High School**, *Academic*, Shanghai Private New Fuxing Junior High School, Shanghai, Shanghai, China

Publications

- OOPSLA'21 **Jialu Zhang**, Ruzica Piskac, Ennan Zhai, Tianyin Xu: "Statically Detecting Silent Misconfigurations with Deep Interaction Analysis"
- ASE'22 **Jialu Zhang**, De Li, John Kolesar, Hanyuan Shi, Ruzica Piskac: "Automated Feedback Generation for Competition-Level Code"
- ISSTA'22 **Jialu Zhang**, Todd Mytkowicz, Mike Kaufman, Ruzica Piskac and Shuvendu Lahiri: "Using Pre-trained Language Models to Resolve Textual and Semantic Merge Conflicts (Experience Paper)"
- SANER'22 Mark Santolucito, **Jialu Zhang**, Ennan Zhai, Jurgen Cito, Ruzica Piskac: "Learning CI Configuration Correctness for Early Build Feedback"
- OOPSLA'24 **Jialu Zhang**, José Cambronero, Sumit Gulwani, Vu Le, Ruzica Piskac, Gustavo Soares, Gust Verbruggen: "PyDex: Repairing Bugs in Introductory Python Assignments using LLMs"

Teaching Experience

Introduction to Computer Structures and Real-time Systems (Spring 2025),
University of Waterloo.

Services

OOPSLA'25 Program Committee

OOPSLA'23 Program Committee

Reviewer ACM Transactions on Software Engineering and Methodology, Empirical Software Engineering, ACM SIGPLAN International Conference on Systems, Programming, Languages and Applications: Software for Humanity, International Conference on Automated Software Engineering (ASE) Industry Showcase

Lab & Talk GAINS: Girls Advancing in STEM, 2022

Judge International Science and Engineering Fair (ISEF), 2024

Honors and Awards

2017-2023 Yale University Graduate Fellowship

2017 A+ Senior Thesis, Shanghai Jiao Tong University (Top 5%)

2016, 2017 Academic Excellence Scholarship of Shanghai Jiao Tong University